

Closing the Marginal-to-Conditional Coverage Gap: Mondrian Conformal Prediction in ER Discrete-Event Simulation

Course: 23AID201

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Abstract

Standard conformal prediction (CP) guarantees marginal coverage - correct on average across a calibration set - but gives no guarantee within any specific subgroup or operating condition. Gopakumar et al. (2026) name this as an explicit, untested limitation of CP for surrogate-model uncertainty quantification, validated only in physics simulation domains. This report tests whether the limitation holds, and whether Mondrian conformal prediction (which calibrates separately per category instead of pooling) closes it, in a discrete-event ER queueing simulation - a domain not previously tested. We find a single pooled quantile silently fails the operationally critical scenario (understaffed, high demand), with coverage as low as 68.2% against a 90% target, while Mondrian CP corrects it to 90.9-92.0%. The effect is statistically significant ($p < 0.001$, 30 independent repeats) and replicates at a second, independent hospital department.

1. Introduction

Surrogate models - fast learned approximations of expensive simulations - are increasingly used to support operational decisions in stochastic service systems such as emergency room (ER) staffing. A point prediction alone is insufficient for such decisions: a claim like "expected wait time is 40 minutes" is far less useful than a calibrated interval such as "35-55 minutes, 90% of the time." Conformal prediction (CP) provides such intervals with a distribution-free, finite-sample coverage guarantee, but standard CP's guarantee is marginal - averaged across the entire calibration distribution. It says nothing about whether coverage holds within a specific subgroup, such as a particular staffing level or demand regime.

Gopakumar et al. (2026) validate CP for surrogate-model uncertainty quantification (UQ) in physics simulation domains (PDEs, magnetohydrodynamics, weather, fusion), and explicitly flag marginal-only coverage as an untested limitation. This report investigates that limitation in a fundamentally different domain - discrete-event queueing simulation of an emergency department - and evaluates Mondrian conformal prediction, which partitions calibration by category to target conditional rather than purely marginal coverage, as a remedy.

2. Method

A discrete-event simulation (DES) of an ER, built in SimPy, was calibrated on real arrival and triage-acuity (ESI) data from a Yale New Haven Health System dataset (department A, the largest/academic site: 322,283 visits, 258.2 visits/day). A gradient-boosting surrogate model was trained to approximate the DES's outputs (patient count, mean wait, mean total time, 95th-percentile wait) as a function of two scenario parameters: staffing capacity and arrival-rate multiplier.

Standard CP calibrates a single quantile of the surrogate's absolute residuals on a held-out calibration set, applied uniformly to every test point. Mondrian CP instead partitions the calibration set into categories - here, the cross of staffing tercile (Low/Medium/High) and arrival-rate tercile (Low/Medium/High), nine cells in total - and calibrates a separate quantile per cell. Category bin edges are derived from the calibration set alone, never the test set. Both methods target $\alpha = 0.1$ (90% nominal coverage) on identical calibration and test data, so the comparison isolates the effect of pooling versus partitioning.

To confirm the finding is not a single-split artifact, the full pipeline was repeated across 30 independent (calibration, test) draws, each freshly generated from the DES, and Mondrian CP's coverage advantage was tested with a paired t-test (all methods share the same 30 draws).

3. Results

Table 1 shows the worst-performing category under standard (pooled) CP for each target, compared to Mondrian CP's coverage in that same category, on a single representative split. The worst category is consistently staff = Low / arrival = High - an understaffed ER during a demand surge, the scenario that matters most operationally.

Target	Pooled coverage (worst category)	Mondrian coverage (same category)	Target coverage
mean_wait_minutes	68.2%	90.9%	90%
mean_total_minutes	80.7%	92.0%	90%
p95_wait_minutes	72.7%	92.0%	90%

Table 1. Coverage in the hardest operating condition under pooled vs. Mondrian calibration. A single pooled quantile, calibrated on average difficulty, silently fails this scenario while simultaneously overcovering the easiest category (staff = High / arrival = Low reaches exactly 100% pooled coverage - wasted interval width).

Repeating the full pipeline across 30 independent calibration/test draws confirms the effect is stable, not a lucky split (Table 2): standard deviations of 1-1.3 percentage points and 95% confidence intervals of 0.35-0.46 points across every target/method combination.

Target	Method	Coverage (mean $\pm$ 95% CI)	Width (mean $\pm$ 95% CI)
mean_wait_minutes	GP baseline	88.31% $\pm$ 0.47%	44.4 $\pm$ 1.1
mean_wait_minutes	Standard CP	90.09% $\pm$ 0.46%	48.0 $\pm$ 1.5
mean_wait_minutes	Mondrian CP	91.07% $\pm$ 0.35%	41.7 $\pm$ 1.4
p95_wait_minutes	GP baseline	88.97% $\pm$ 0.44%	354.6 $\pm$ 11.5
p95_wait_minutes	Standard CP	90.06% $\pm$ 0.39%	377.1 $\pm$ 12.6
p95_wait_minutes	Mondrian CP	91.35% $\pm$ 0.46%	328.6 $\pm$ 13.4

Table 2. 30-repeat statistical summary. Mondrian CP's mean coverage sits consistently above 90% across every target, while a Gaussian Process (GP) baseline consistently undercovers and standard CP lands close to but sometimes under target. Mondrian CP is also narrower than standard CP on three of four

targets, indicating pooled calibration was wasting width in easy categories to compensate for hard ones, not simply redistributing it.

A paired t-test on per-repeat coverage (all methods share the same 30 draws) confirms Mondrian CP's advantage over both standard CP and the GP baseline is statistically significant at $p < 0.001$ on every one of the four targets (e.g. $p = 1.42e-10$ for Mondrian CP vs. GP on `mean_wait_minutes`).

`n_patients` is the one target where Mondrian CP does not help: its pooled coverage was already fairly uniform across categories (87-99%), so there was no real conditional miscalibration to correct, and splitting calibration into nine smaller cells (~130 points each vs. 1,200 pooled) adds finite-sample noise without a corresponding benefit. This is a known, expected Mondrian CP tradeoff, and a more credible finding than uniform improvement everywhere would have been.

The finding replicates at Department B, a second, independent site in the same dataset with roughly half the patient volume and a meaningfully different acuity mix (a community rather than academic ED): the same worst category (understaffed, high demand) shows pooled coverage of 76.2-81.0% there, corrected to 86.9-90.5% by Mondrian CP - closely matching the magnitude of correction seen at Department A.

Combining Mondrian CP's per-category calibration with an adaptive, quantile-regression-based nonconformity score (Mondrian CQR) improves further on the hardest target: for `p95_wait_minutes`, Mondrian CQR achieves both higher coverage (+0.82 percentage points, $p = 0.003$) and narrower intervals (-35.9, $p < 1e-17$) than Mondrian CP alone, suggesting the two techniques address partially distinct sources of miscalibration.

4. Discussion

The result demonstrates concretely why marginal coverage is an insufficient guarantee for operational decision-making: a method that is "correct on average" can be systematically wrong in exactly the situations where a decision-maker most needs it to be right. In an ER staffing context, this is precisely the surge scenario where an unreliable uncertainty estimate carries the highest cost. Mondrian CP's category-conditional calibration directly addresses this by giving each operating regime its own, honestly-sized interval, at a modest cost in per-category calibration sample size.

This is, to our knowledge, the first empirical test of conformal prediction's marginal-coverage limitation outside the physics simulation domains in which it was originally validated. The result - that the limitation is real and that Mondrian CP closes it where a genuine conditional gap exists - extends Gopakumar et al.'s findings to a discrete-event, queueing-based domain.

5. Conclusion

Standard conformal prediction's marginal coverage guarantee masks a real and operationally important failure mode in ER queueing surrogate models: silent undercoverage in the understaffed, high-demand regime. Mondrian conformal prediction corrects this failure, with the correction validated by statistical significance testing across 30 independent repeats and replicated at an independent hospital department. The one target where Mondrian CP does not help is explained by an absence of real conditional miscalibration, not hidden as a negative result.

References

Gopakumar, et al. (2026). Conformal Prediction for surrogate-model uncertainty quantification in physics simulation domains (PDEs, MHD, weather, fusion). [Base paper]

Hospital Triage and Patient History Data. Kaggle (maalona) - Yale New Haven Health System, retrospective study, March 2014 - July 2017. [Real-world calibration data]